

# Yousef Mohammad

Software Engineer · Backend & Full-Stack · Applied AI

Växjö, Sweden

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Software engineer with experience in TypeScript/Python full-stack platforms and C#.NET desktop applications. Background in backend development, SQL/NoSQL databases, applied AI/RAG and Docker-based CI/CD delivery.

## SKILLS

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|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages</b>            | C#, Python, TypeScript, JavaScript, Java, SQL, C++, Dart, GLSL, Bash                                                                 |
| <b>.NET &amp; Desktop</b>   | .NET 10, Avalonia UI, Microsoft.Data.Sqlite, QuestPDF, xUnit, dependency injection, repository interfaces                            |
| <b>Backend &amp; Web</b>    | FastAPI, Flask, Node/Nitro, REST APIs, Vue 3, Nuxt 3, React, authentication, RBAC, SSE, D3.js, Chart.js, Three.js                    |
| <b>AI &amp; Data</b>        | RAG, LLM integration, prompt/context engineering, embeddings, vector databases, OpenAI API, BioBERT, PubMedCLIP, MLflow, data mining |
| <b>Databases</b>            | SQLite, MongoDB, PostgreSQL, MySQL, data modelling, analytics, SQL and NoSQL workflows                                               |
| <b>Cloud &amp; Platform</b> | Linux/Unix, Docker, Compose/Swarm, Kubernetes basics, Git, GitHub/GitLab CI/CD, Kaniko, Nginx, AWS S3, Cloudinary, Prefect           |
| <b>Systems</b>              | CUDA optimisation, OpenGL, compiler design, embedded/robotic systems, memory management and low-level debugging                      |

## EXPERIENCE

Research Engineer / Full-Stack Developer [Linnaeus University](#)

Växjö, Sweden

Jun 2024 to Jul 2026

### AWAITS

- Co-developed AI-supported educational software used by approximately 150 students for academic writing and feedback workflows.
- Built and maintained Nuxt 3/TypeScript and MongoDB features including rich-text editing, review flows, teacher dashboards, user management, consent handling, analytics and role-based access.
- Delivered the platform with Docker, GitLab CI/CD, Kaniko, Docker Swarm, health checks and load testing; co-authored the resulting peer-reviewed conference paper.

### InKuiS

- Built a centralised platform for discovering local events, with automated ingestion, validation and categorisation flows plus LLM-supported open-question search.
- Implemented authentication, profiles, uploads, calendar and analytics views, bilingual UI, and location search with geocoding, radius filtering and distance sorting.

### Robotics

- Integrated an LLM API with a Misty Robotics robot through HTTP/WebSocket communication, enabling real-time conversational interaction in K–12 classroom activities.

### Medical Visualisation and Analytics

- Built a full-stack tool for exploring 7,400+ medical cases and 40,900+ images through UMAP views, linked filters, dashboards and text/image-embedding retrieval using FastAPI, BioBERT and PubMedCLIP.

## TECHNICAL PROJECTS

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### TimeReady – Employee Scheduling and Payroll Preparation

- Built a C#/.NET 10 and Avalonia desktop application for scheduling, clock-in/out and attendance review, separating domain models, application services, SQLite persistence and presentation.
- Implemented constructor dependency injection, repository interfaces, transactional storage, PIN hashing and role-based access, with audited payroll approval, locking and CSV export.
- Added xUnit tests for scheduling rules, persistence, migrations, authentication and payroll workflows; configured GitHub Actions build/test checks and Windows installer packaging.

### Faktura Generator – Swedish Invoicing Application

- Built a C#/.NET 10 and Avalonia application with invoice validation, VAT calculations and QuestPDF generation, plus asynchronous PDF file writes.
- Implemented a SQLite repository with transactions, schema migrations, invoice/customer numbering and revenue dashboards; configured GitHub Actions packaging for Windows, macOS and Linux.

### Java Compiler and Language Tooling

- Implemented ANTLR parsing, symbol tables, scope and type checking, Python code generation and JVM bytecode generation with ASM.

## EDUCATION

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**M.Sc. in Software Engineering, 300 ECTS** [Linnaeus University](#)

Växjö, Sweden  
2021 to 2026

- **Master's thesis – RiskRank:** Context-aware prioritisation of CI/CD vulnerabilities using scanner findings, fan-in ranking, git-history signals and reviewer-focused reports.
- **Bachelor's thesis:** Built and comparatively evaluated a RAG-based academic writing assistant using retrieval, prompting and vector databases.
- Selected work: CUDA matrix-multiplication optimisation, OpenGL game-engine development, compiler construction, machine/deep learning, computer vision, mathematics and physics.

## APPLIED AI & RETRIEVAL

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- Built and evaluated RAG systems using embeddings, vector databases and prompting; integrated LLMs into deployed academic-writing workflows and event search.
- Implemented multimodal retrieval with BioBERT and PubMedCLIP; combined text/image search with interactive visual analytics.

## PUBLICATIONS & RECOGNITION

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- **AWAITS:** Co-author of a peer-reviewed conference paper on a digital system for academic writing support; the project received Innovation Call recognition in 2024.
- **Misty Robotics:** Classroom robotics work was featured by Misty Robotics.

## LANGUAGES

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Swedish (native) · English (full professional proficiency) · Arabic (native)